EchoST-SAM: Semantic-Prompted Spatiotemporal LV Segmentation for Functional Analysis in Echocardiography

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***Abstract*—Speckle noise and weak endocardial boundaries make frame-wise left-ventricular (LV) segmentation in cine echocardiography unstable, potentially affecting measurements derived from time-varying contour geometry. We propose EchoST-SAM, a semantic-prompted framework for interaction-free spatiotemporal LV segmentation and mask-derived geometric assessment. An adapted SAM Vision Transformer and a local feature-enhancement branch capture complementary global and local representations. Fixed anatomical phrases encoded by PubMedBERT provide a shared semantic prior for automatic box prompting through Grounding DINO and prompt-aware visual feature weighting. ConvLSTM propagates the fused features across consecutive frames before mask decoding. Evaluated on CAMUS A2C sequences with clinician-provided intermediate-frame annotations, EchoST-SAM achieves 93.90% Dice, a 9.11-pixel Hausdorff distance, and a temporal consistency of 0.956. Compared with its frame-wise spatial variant, the recurrent model increases temporal consistency by 0.035 and reduces the mean absolute errors of ellipse eccentricity and fractional cavity-area change from 0.0142 to 0.0131 and from 0.071 to 0.064, respectively, with a 0.07-percentage-point Dice decrease. These descriptive results suggest a trade-off between frame-level accuracy and temporal consistency with improved geometric readouts on CAMUS A2C. Functional metrics are evaluated at reference end-diastolic/end-systolic frames to isolate segmentation-derived measurement errors.**

***Index Terms*—echocardiography, functional analysis, left ventricular segmentation, semantic prompting, spatiotemporal segmentation**

I. Introduction

Echocardiography is a low-cost, radiation-free modality for observing cardiac anatomy and motion, but its quantitative analysis remains labor intensive. Automated delineation of the left ventricular (LV) endocardium at end diastole (ED) and end systole (ES) can facilitate chamber volume estimation and systolic function assessment [1]. However, accurate segmentation remains challenging because ultrasound images contain speckle, signal dropout, low contrast, and view-dependent artifacts [2]. Even a localized contour error near the apex or mitral plane can affect the estimated cavity geometry and subsequent functional measurements.

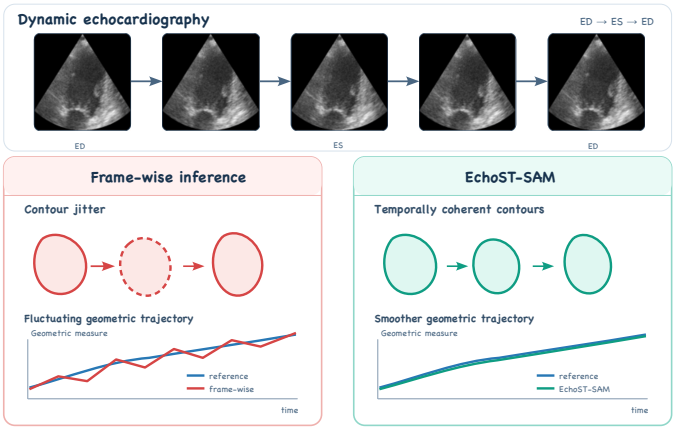


Fig. 1. Spatiotemporal LV segmentation across the ED–ES–ED cardiac cycle. Frame-wise inference may produce contour jitter and fluctuating geometric trajectories, whereas EchoST-SAM propagates temporal features to promote temporally coherent contours and smoother mask-derived geometric trajectories. The trajectories are conceptual illustrations.

Convolutional and Transformer-based models provide strong frame-level segmentation baselines [3]–[5]. More recently, the Segment Anything Model (SAM) introduced promptable segmentation through large-scale pretraining [6]. However, direct transfer of SAM to medical images remains challenging [7]. MedSAM [8], SAMed [9], SAMUS [10], and CC-SAM [11] therefore explore domain-specific training, parameter-efficient adaptation, or architectural modifications. Despite these advances, two practical limitations remain relevant to cine echocardiography. First, methods requiring manual clicks or boxes need prompt automation to support an interaction-free full-cycle segmentation workflow. Second, independent frame processing ignores temporal dependencies and may produce inconsistent contours despite high average Dice.

Temporal inconsistency is not merely a visual artifact. Cine echocardiography records a structured contraction–relaxation process, and unstable contours can distort measurements computed across time, as illustrated in Fig. 1. Video-based models

have demonstrated the value of cardiac motion for functional analysis [12], while temporal correction can improve segmentation consistency [13]. MediViSTA further introduces temporal adaptation into SAM-based echocardiographic video segmentation [14]. These developments motivate integrating automatic prompting, spatial adaptation, and temporal modeling. A key question is how to combine anatomical guidance, local boundary information, temporal feature propagation, and traceable geometric measurements within a unified LV cine segmentation framework.

To address this question, we propose **EchoST-SAM**, a semantic-prompted framework for interaction-free spatiotemporal LV segmentation and mask-derived geometric assessment. Its spatial encoder combines adapted global SAM features with local ultrasound boundary information. Anatomical descriptions at multiple granularities provide a shared semantic prior for frame-specific box prompting and visual feature weighting. ConvLSTM then propagates the fused semantic-spatial features over consecutive frames before mask decoding. The resulting masks yield ellipse eccentricity and fractional cavity-area change as traceable geometric readouts. For full-cycle evaluation, we supplement the official CAMUS ED/ES annotations with clinician-provided LV masks for intermediate A2C frames; ED/ES-based measurement evaluation uses reference frame indices. Our main contributions are:

- An interaction-free SAM adaptation combines global–local visual features with anatomical guidance for automatic box prompting and feature weighting, supporting target localization and weak endocardial boundary delineation.
- ConvLSTM propagates the fused semantic-spatial features to improve the reported temporal consistency and geometric errors with a small frame-level Dice trade-off while retaining the SAM decoding interface.
- A traceable geometric readout and dense full-cycle evaluation jointly quantify overlap, boundary accuracy, temporal consistency, and two complementary measurements: ellipse eccentricity and fractional cavity-area change.

II. Related Work

A. LV segmentation in echocardiography

CAMUS provides a public benchmark for chamber segmentation and functional analysis in heterogeneous 2D echocardiography [15]. U-Net and its variants effectively capture local appearance [3], while attention-based architectures, including TransUNet, Swin-Unet, MISSFormer, and H2Former, model broader spatial context [4], [5], [16], [17]. However, processing frames independently does not enforce temporal coherence. Painchaud et al. used an interpretable shape space to identify and correct temporal inconsistencies [13], whereas Ouyang et al. demonstrated the value of full cardiac videos for beat-to-beat assessment [12]. Xu et al. explored multi-task self-supervised learning for automated measurement of left ventricular ejection fraction in echocardiography [18].

B. Foundation models and semantic prompting

SAM provides a general promptable segmentation interface learned from large-scale natural images [6]. Medical adaptations have explored domain-specific training [8], low-rank fine-tuning [9], lightweight adapters [19], and hybrid CNN–Transformer encoders for ultrasound [10], [11]. Recent evidence further indicates that the relative benefit of foundation and task-specific models depends on the cardiac ultrasound setting [20]. Language models provide complementary anatomical priors: PubMedBERT encodes biomedical text [21], Grounding DINO localizes phrase-referred objects [22], and TextSAM-EUS demonstrates automatic text-guided ultrasound segmentation [23]. Deng et al. further explored LLM-guided SAM for LV segmentation and functional analysis in echocardiography [24].

C. Temporal Modeling for Video Segmentation

ConvLSTM models temporal dependencies through convolutional recurrent operations that preserve spatial structure [25]. Space-time attention can capture longer-range interactions [26], but typically incurs greater computational and data demands. For general promptable video segmentation, SAM-I2V extends SAM with spatiotemporal feature extraction, memory filtering, and memory-based prompting to support temporally consistent mask propagation [27]. In echocardiography, MediViSTA introduces temporal adapters into SAM for video segmentation [14].

III. Methodology

A. Problem Formulation and Overview

Given an echocardiographic cine $\mathcal{I} = \{I_t\}_{t=1}^{T_c}$ with binary LV masks $\mathcal{Y} = \{Y_t\}_{t=1}^{T_c}$, EchoST-SAM predicts $\hat{\mathcal{Y}}$ without user-provided prompts. As shown in Fig. 2, anatomical text guides automatic box generation and global–local feature weighting. ConvLSTM propagates the resulting features across frames, and the SAM decoder predicts masks for geometric assessment.

B. Semantic-Prompted Spatial Encoding

1) Global–local visual representation: Each 256×256 frame is processed by overlapping patch embedding and a frozen SAM ViT with trainable feature and position adapters, following SAMUS [10], [28]. The feature adapter is added residually:

$$A(x) = \mathrm{GELU}(xP_d)P_u, \tag{1}$$

where P_d and P_u reduce and restore channel dimensions. A parallel LFE branch uses downsampling and dilated/standard 3×3 convolutions to capture local appearance. Both branches produce 32×32 feature maps.

After projection to width d, CBA fuses global and local tokens $F_t^v, F_t^\ell \in \mathbb{R}^{N_s \times d}$, with $N_s = 1024$:

$$\tilde{F}_t = F_t^v + \mathrm{softmax}\left(\frac{(F_t^v W_q)(F_t^\ell W_k)^\top}{\sqrt{d}} + R\right)(F_t^\ell W_v), \tag{2}$$

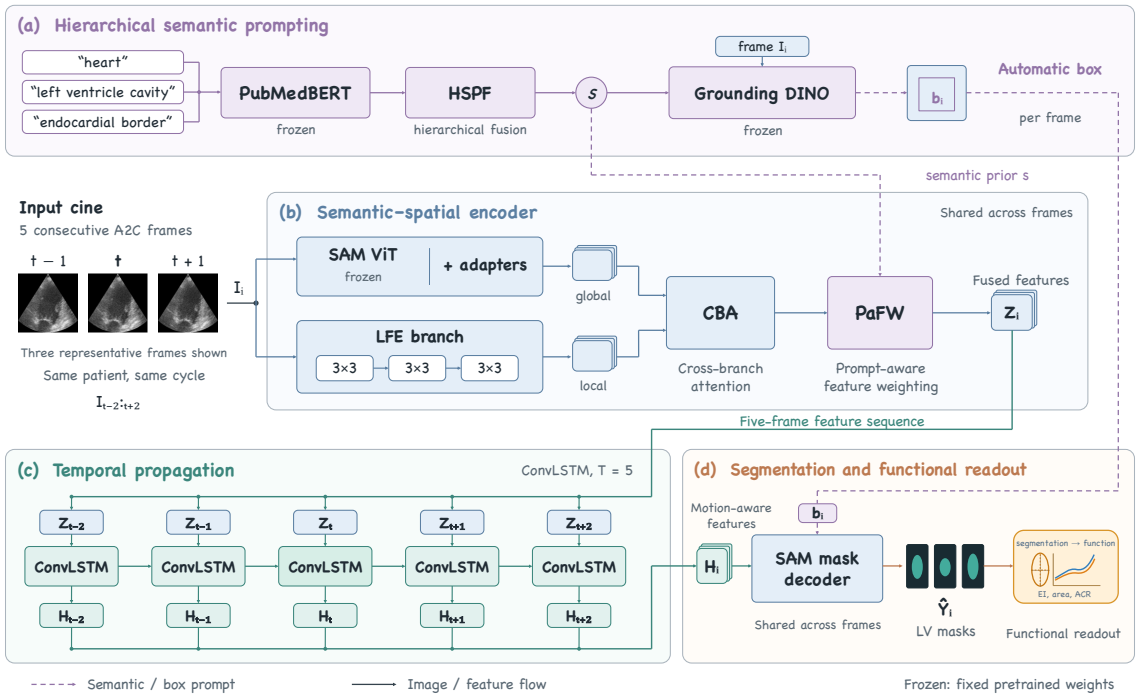


Fig. 2. Overview of EchoST-SAM. PubMedBERT and HSPF provide anatomical conditioning for automatic box prompting and prompt-aware feature weighting (PaFW). Cross-branch attention (CBA) integrates adapted SAM and local feature-enhancement (LFE) representations. ConvLSTM propagates the fused features over five-frame clips before SAM mask decoding and EI/ACR calculation.

where W_q, W_k, W_v are learned projections and R is a relative-position bias. Global tokens act as queries, while local tokens provide keys and values.

2) Hierarchical anatomical prompting: Frozen PubMedBERT encodes three fixed phrases, *heart*, *left ventricle cavity*, and *endocardial border*, as $e_n = E_{\text{BERT}}(p_n)$. HSPF pools and projects them:

$$e = W_2\,\text{GELU}(W_1\bar{e} + b_1) + b_2, \qquad \bar{e} = \frac{1}{N}\sum_{n=1}^{N} e_n, \tag{3}$$

where $N = 3$ and the projection parameters are trainable. The cached text embeddings yield a shared, image-independent prior. Here, "hierarchical" refers to anatomical granularity; mean pooling does not explicitly model hierarchical relations.

The box-prompt pathway is expressed as

$$b_t = G_{\text{DINO}}(I_t, e), \qquad s_t = E_{\text{prompt}}(b_t), \tag{4}$$

where frozen Grounding DINO [22] generates a frame-specific box b_t, and the frozen SAM prompt encoder maps it to the prompt embedding s_t.

3) Prompt-aware feature weighting: PaFW weights each visual token $\tilde{f}_{t,j}$ according to its cosine alignment with the projected semantic prior:

$$a_{t,j} = \frac{\tilde{f}_{t,j}^{\top} W_e e}{\|\tilde{f}_{t,j}\|_2 \|W_e e\|_2}, \qquad w_{t,j} = \frac{\exp(a_{t,j})}{\sum_{k=1}^{N_s}\exp(a_{t,k})}, \qquad Z_{t,j} = w_{t,j}\tilde{f}_{t,j}. \tag{5}$$

Weights are normalized across spatial tokens, and the weighted features are reshaped into 32×32 maps for temporal propagation.

C. Temporal Feature Propagation

ConvLSTM [25] propagates the weighted features while preserving their spatial layout:

$$(H_t, C_t) = \text{ConvLSTM}(Z_t, H_{t-1}, C_{t-1}), \tag{6}$$

where H_t and C_t denote the hidden and memory states. The adapted SAM decoder combines H_t with the box embedding:

$$\hat{Y}_t = M_{\text{dec}}(H_t, s_t). \tag{7}$$

We use chronological clips of length $T = 5$, initialize both states to zero for each clip, and share the decoder across frames. Training and inference preserve frame order; predictions are mapped back to their original indices for cycle-level evaluation.

D. Training Objective

All frames receive supervision through binary cross-entropy, soft Dice loss, and PaFW entropy regularization:

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T}\Big[\mathcal{L}_{\text{BCE}}(\hat{Y}_t, Y_t) + \mathcal{L}_{\text{Dice}}(\hat{Y}_t, Y_t) + \lambda_{\text{pfw}}\mathcal{H}(w_t)\Big],$$
$$\mathcal{H}(w_t) = -\sum_{j} w_{t,j}\log w_{t,j}, \tag{8}$$

with $\lambda_{\mathrm{pfw}} = 0.1$. Minimizing the entropy term encourages concentrated token weights. We train the adapters, LFE, HSPF, PaFW, ConvLSTM, and decoder, while freezing the pretrained ViT, PubMedBERT, Grounding DINO, and SAM prompt encoder.

E. Segmentation-Derived Functional Readout

The foreground-pixel count gives area A_t, while an ellipse fitted to the largest contour provides major and minor axes $L_{t,\mathrm{maj}}$ and $L_{t,\mathrm{min}}$. We calculate

$$\mathrm{EI}_t = \sqrt{1 - \left(\frac{L_{t,\mathrm{min}}}{L_{t,\mathrm{maj}}}\right)^2}, \qquad \mathrm{ACR} = \frac{A_{\mathrm{ED}} - A_{\mathrm{ES}}}{A_{\mathrm{ED}}}. \quad (9)$$

EI measures ellipse elongation, rather than the clinical short-axis eccentricity ratio; ACR measures fractional A2C cavity-area change. For visualization, $\mathrm{ACR}_t = (A_{\mathrm{ED}} - A_t)/A_{\mathrm{ED}}$.

Predicted area maxima and minima provide a possible ED/ES selection rule, but the reported evaluation uses reference phase indices to isolate segmentation-derived error. It therefore does not validate automatic phase detection or complete functional trajectories. Both measurements are geometric surrogates, not substitutes for biplane LVEF or clinical diagnosis.

IV. Experiments

A. Dataset

We use the A2C recordings of 500 CAMUS patients with official ED/ES annotations [15], supplemented by quality-checked clinician LV masks for all intermediate frames. A fixed, study-specific patient-disjoint split assigns 400/50/50 patients to training/validation/testing. All methods use the same split, dense annotations, and 256×256 resolution. Inputs are chronological clips from the same patient and cycle, with $T = 5$ denoting the default window length rather than cine length.

B. Implementation Details

EchoST-SAM is trained for 200 epochs on one NVIDIA RTX 4090 using Adam with batch size 16, learning rate 10^{-2}, and weight decay 10^{-4}. StepLR reduces the learning rate by a factor of 0.1 every 50 epochs. Validation determines checkpoints and window length; the test set is reserved for evaluation. Trainable and frozen components follow Section III. External baselines use their respective adaptation settings under the same data and evaluation protocol.

C. Evaluation Metrics

We report Dice, symmetric maximum Hausdorff distance (HD, pixels), EI/ACR mean absolute errors (MAEs), and temporal consistency:

$$\mathrm{TC} = \frac{1}{T_c - 1} \sum_{t=1}^{T_c - 1} \mathrm{Dice}(\hat{Y}_t, \hat{Y}_{t+1}), \quad (10)$$

averaged across test cines. EI/ACR MAEs compare predicted and annotation-derived geometry, with ACR evaluated at reference ED/ES indices. Since TC may reward over-smoothing, we interpret it jointly with segmentation accuracy and geometric errors. Agreement is assessed using Pearson correlation, mean bias, and 95% Bland–Altman limits of agreement.

D. Quantitative Comparison

We compare EchoST-SAM with six task-specific models: U-Net [3], Swin-Unet [5], MISSFormer [16], TransUNet [4], AAU-Net [29], and H2Former [17]; and four SAM-based models: MedSAM [8], SAMed [9], SAMUS [10], and CC-SAM [11].

TABLE I
SEGMENTATION AND GEOMETRIC ASSESSMENT ON CAMUS A2C. BOLD/UNDERLINED VALUES INDICATE THE BEST/SECOND-BEST RESULTS. THE SPATIAL VARIANT EXCLUDES CONVLSTM.

Method	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
Task-specific methods				
U-Net	93.56	9.90	0.0167	0.082
Swin-Unet	91.88	13.11	0.0192	0.087
MISSFormer	93.25	9.94	0.0163	0.084
TransUNet	93.60	9.60	0.0155	0.080
AAU-Net	93.59	9.13	0.0160	0.078
H2Former	93.44	9.60	0.0153	0.077
SAM-based methods				
MedSAM	87.52	15.28	0.0239	0.091
SAMed	87.67	13.24	0.0242	0.094
SAMUS	91.13	11.76	0.0172	0.079
CC-SAM	92.85	10.88	0.0150	0.074
Proposed methods				
EchoST-SAM (spatial)	**93.97**	**9.05**	<u>0.0142</u>	<u>0.071</u>
EchoST-SAM	<u>93.90</u>	<u>9.11</u>	**0.0131**	**0.064**

As shown in Table I, EchoST-SAM exceeds CC-SAM, the strongest external SAM-based baseline in this comparison, by 1.05 Dice percentage points and reduces EI/ACR MAEs by 12.7%/13.5%. Compared with its spatial variant, it reduces EI/ACR MAEs by 7.7%/9.9%, with a 0.07-percentage-point Dice decrease and a 0.06-pixel HD increase. The recurrent model achieves the lowest geometric errors, whereas the spatial variant retains the highest Dice and lowest HD. These point estimates suggest a trade-off between frame-level segmentation accuracy and mask-derived geometric measurements; statistical significance remains unestablished.

E. Qualitative Analysis

Fig. 3 compares EchoST-SAM (spatial) and the full EchoST-SAM model over five consecutive frames. In this example, the full model shows visually smoother contour evolution around the apex and lateral wall, consistent with its higher reported TC. Magnified regions from the third frame facilitate inspection of local boundary differences. This illustrative comparison does not establish physiological fidelity or general robustness.

Potential failure modes include persistent endocardial boundary dropout, apical truncation, and confusion with adjacent chambers. Temporal recurrence may propagate localization errors across frames, while semantic prompting cannot recover anatomy absent from the acquisition. Future work

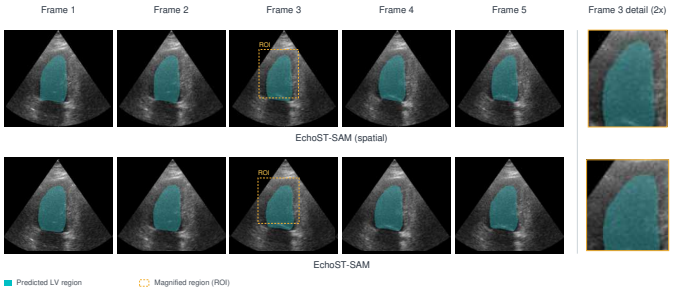


Fig. 3. LV predictions over five consecutive frames from EchoST-SAM (spatial) (top) and EchoST-SAM (bottom). Cyan denotes the predicted LV region. Yellow dashed boxes indicate regions in the third frame displayed at $2\times$ magnification on the right.

could investigate uncertainty-aware state updates to reduce the influence of unreliable frames during temporal propagation.

F. Ablation Studies

1) Spatial components: Table II evaluates the spatial components with ConvLSTM disabled. Removing LFE produces the largest Dice decrease (1.37 percentage points) and HD increase (0.97 pixels), while removing semantic prompting, HSPF, or PaFW also worsens all four metrics. The full spatial model achieves the best reported results, suggesting that each component contributes under the evaluated configuration.

TABLE II
SPATIAL COMPONENT ABLATIONS ON CAMUS A2C. CONVLSTM IS DISABLED IN ALL VARIANTS; BOLD INDICATES THE BEST RESULTS.

Variant	Dice (%) ↑	HD (px) ↓	EI MAE ↓	ACR MAE ↓
w/o LFE	92.60	10.02	0.0167	0.083
w/o semantic prompting	92.84	9.88	0.0162	0.081
w/o HSPF	93.08	9.63	0.0154	0.079
w/o PaFW	93.21	9.52	0.0150	0.078
Full spatial model	**93.97**	**9.05**	**0.0142**	**0.071**

These observations are consistent with the intended roles of local feature extraction, anatomical conditioning, and feature weighting, but do not directly establish their mechanisms or statistical significance. In particular, removing semantic components does not distinguish the benefit of biomedical semantics from that of a shared learned prior; a matched non-semantic control is needed.

2) Temporal window length: Table III compares temporal window lengths. Increasing T from 1 to 5 raises TC from 0.921 to 0.956, with a 0.07-percentage-point Dice decrease and a 0.06-pixel HD increase. Extending the window to $T = 7$ adds only 0.001 TC while further reducing Dice and increasing HD. The default $T = 5$ follows the validation-based selection protocol described above.

The results illustrate a trade-off between adjacent-mask consistency and frame-level accuracy: longer windows do not uniformly improve all metrics. Although $T = 5$ provides higher Dice and lower HD than the other evaluated multi-frame settings, the small differences remain descriptive without uncertainty estimates. The results alone cannot determine whether the degradation at $T = 7$ reflects error propagation, optimization variability, or another factor.

TABLE III
EFFECT OF TEMPORAL WINDOW LENGTH ON CAMUS A2C. BOLD/UNDERLINED VALUES INDICATE THE BEST/SECOND-BEST RESULTS; $T = 5$ IS THE DEFAULT SETTING.

Window T (frames)	Dice (%) ↑	HD (px) ↓	TC ↑
1	**93.97**	**9.05**	0.921
3	93.84	9.28	0.944
5	<u>93.90</u>	<u>9.11</u>	<u>0.956</u>
7	93.76	9.33	**0.957**

G. Functional Assessment and Agreement

Mask-derived EI and ACR remain traceable to contour geometry. Pearson and Bland–Altman analysis [30] show that temporal modeling raises EI/ACR correlations from 0.924/0.931 to 0.937/0.945 and reduces mean biases from $+0.003/-0.005$ to $+0.002/-0.002$.

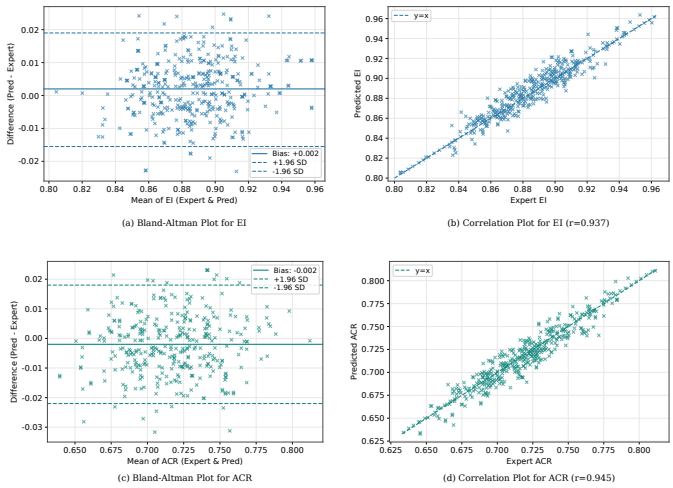


Fig. 4. Agreement between predicted and expert-derived EI (a,b) and ACR (c,d), assessed using Bland–Altman plots (a,c) and correlation plots (b,d). Solid and dashed horizontal lines indicate the mean bias and 95% limits of agreement, respectively; diagonal dashed lines indicate identity ($y = x$).

In Fig. 4, differences center near zero and mostly fall within the 95% limits, consistent with the reported small biases. Together with TC and MAE, these observations suggest improved geometric readouts, but nominal 95% limits do not establish clinically acceptable agreement or rule out over-smoothing.

Correlation and agreement answer different questions here. Higher Pearson correlation indicates stronger linear association, whereas bias and limits reveal systematic and observation-level deviations that correlation alone can conceal. Reporting both guards against interpreting a strong linear association as measurement agreement. The remaining dispersion describes variation among observations but does not by itself exclude regression toward a population mean. EI and ACR

remain geometric surrogates rather than clinical endpoints; two-dimensional ACR is not biplane LVEF. These are single-run descriptive results: uncertainty should be assessed at the patient level because repeated frames are dependent. Ground-truth TC, temporal baselines, automatic-phase evaluation, and external validation remain necessary.

V. Conclusion

We presented EchoST-SAM, an automatic framework integrating biomedical semantics, global–local SAM features, and ConvLSTM propagation for spatiotemporal LV segmentation and mask-derived geometric assessment. On CAMUS A2C, it achieves 93.90% Dice, a 9.11-pixel HD, and a TC of 0.956; temporal modeling improves TC by 0.035 and reduces EI/ACR MAEs to 0.0131/0.064, with a 0.07-percentage-point Dice decrease. These descriptive results suggest improved temporal consistency and geometry, while statistical superiority and generalizability require further validation.

Acknowledgments

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